AMIODARONE

High-alert drug
Thyroid, lung, liver toxicity

CHEMICAL NAME (2-butyl-1-benzofuran-3-yl){4-[2-(diethylamino)ethoxy]-3,5-diiodophenyl}methanone (hydrochloride)
CLASS Class III antiarrhythmic (multichannel blocker)
TRADE NAMES Cordarone · Pacerone · Nexterone (IV) · Aratac

THERAPEUTIC USES / INDICATIONS

- Cardiac arrest (shock-refractory VF/pulseless VT).
- Ventricular tachycardia (stable) and prevention of life-threatening ventricular arrhythmias.
- Atrial fibrillation/flutter; rhythm control and rate control (incl. in heart failure).
- SVT resistant to other drugs.

ADMINISTRATION

- **IV:** dilute in 5% dextrose only; give via central line when possible (peripheral → phlebitis, necrosis); use in-line filter if required by product.
- Cardiac arrest: 300 mg IV/IO bolus (prefilled syringe), no dilution delay.
- **PO** with food; long loading phase.
- Continuous ECG monitoring with IV use.

DOSAGES

- **Cardiac arrest:** 300 mg IV after 3rd shock, then 150 mg after 5th shock.
- **Arrhythmia (IV):** 300 mg (5 mg/kg) over 20–60 min, then 900 mg over 24 h.
- **PO loading:** 200 mg three times daily for 1 week, twice daily for 1 week, then maintenance 100–200 mg daily.
- **Children (arrest):** 5 mg/kg.

PHARMACOKINETICS

- **Onset:** IV hours; PO days to weeks. Very large Vd (lipophilic — stored in fat, liver, lung, skin).
- Metabolised in liver (CYP3A4) to active desethylamiodarone; inhibits CYP2C9, 2D6, 3A4, P-gp.
- Excreted in bile. **t½:** ~50 days (effects persist months after stopping).

NURSING MANAGEMENT

- IV: continuous ECG, BP; check cannula site frequently; use D5W.
- Baseline and 6-monthly TFTs, LFTs, chest X-ray, ECG; eye exam.
- Report new cough, breathlessness, fever (pneumonitis).
- Check INR and digoxin levels when started.
- Monitor HR (hold if <50) and QTc.

PHARMACODYNAMICS

- Mainly blocks **K^+ channels** → prolongs action potential and refractory period (class III).
- Also blocks Na^+ (I), β-receptors (II) and Ca^{2+} channels (IV).
- Contains iodine (37% by weight) → thyroid effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, anorexia, constipation.
- Photosensitivity, blue-grey skin discoloration.
- Corneal microdeposits (halos at night); tremor; bradycardia; hypotension (IV).

ADVERSE EFFECTS (SERIOUS)

- Pulmonary toxicity (pneumonitis, fibrosis) — can be fatal.
- Hypothyroidism or hyperthyroidism.
- Hepatotoxicity; QT prolongation and torsades; heart block.
- Optic neuropathy; peripheral neuropathy; phlebitis (IV).

CONTRAINDICATIONS & CAUTIONS

- Sinus bradycardia, SA block, 2nd/3rd-degree AV block without pacemaker.
- Thyroid dysfunction; iodine hypersensitivity.
- Long QT; use with other QT-prolonging drugs.
- Pregnancy and breastfeeding (except life-threatening cases).

KEY INTERACTIONS

- **Warfarin:** ↑↑ INR — reduce warfarin dose by ~30–50%.
- **Digoxin:** ↑ levels; halve digoxin dose.
- **Simvastatin:** max 20 mg (myopathy).
- **QT drugs** (macrolides, fluoroquinolones, antipsychotics): torsades; **β-blockers, verapamil:** bradycardia.

PATIENT EDUCATION

- Use high-factor sunscreen and cover skin; even weeks after stopping.
- Report cough, breathlessness, visual changes, weight change, palpitations.
- Keep regular blood tests and eye checks.
- Tell every doctor you take amiodarone (many interactions).

KEY POINTS

- Class III multichannel antiarrhythmic.
- Toxicity: lungs, thyroid, liver, eyes, skin.
- Arrest: 300 mg after 3rd shock.
- Very long t½; raises INR and digoxin.